

When to listen. How to transmit. How to keep service.

Joint sensing, robust beamforming and causal scheduling
for terrestrial / non-terrestrial coexistence

Information

Protection

Service

Theory + reproducible finite-model and fresh ray-tracing evidence

A better estimate is useful only if it earns back its cost

Listen longer

Potentially better information

More lost DL/UL resources

The result arrives later

Transmit now

More immediate service

Older channel information

Possible power backoff

Stay silent

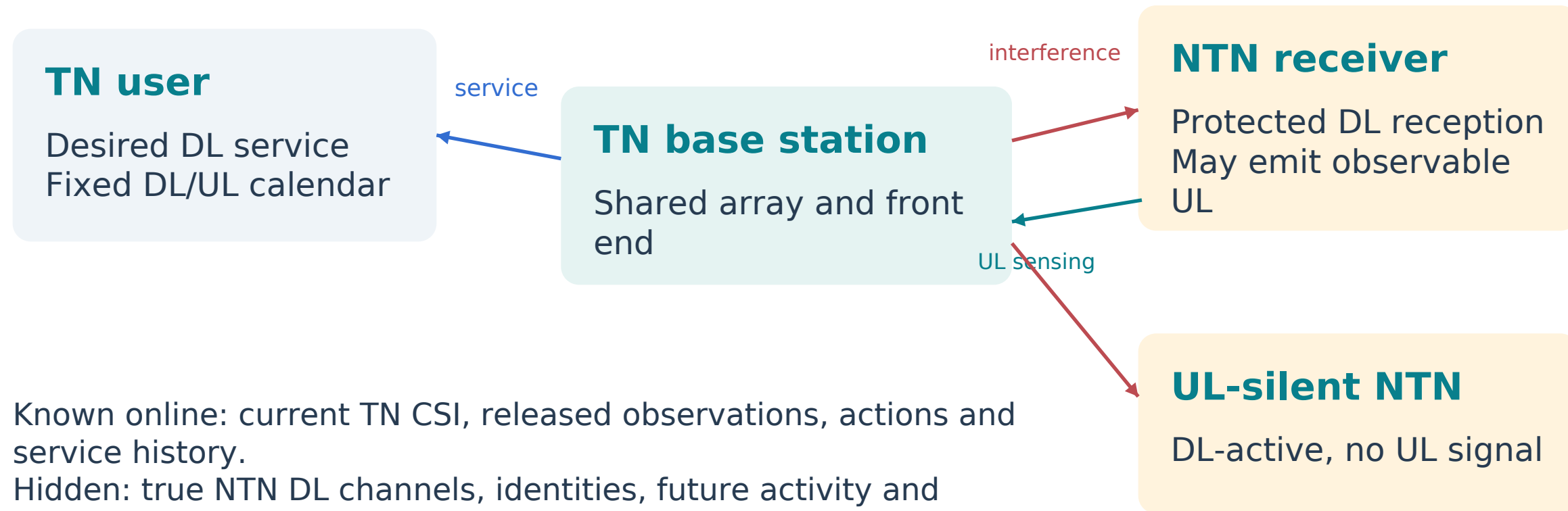
Zero controlled-BS leakage

No TN downlink service

TN uplink may continue

The objective is delivered service after sensing overhead—not estimation accuracy alone.

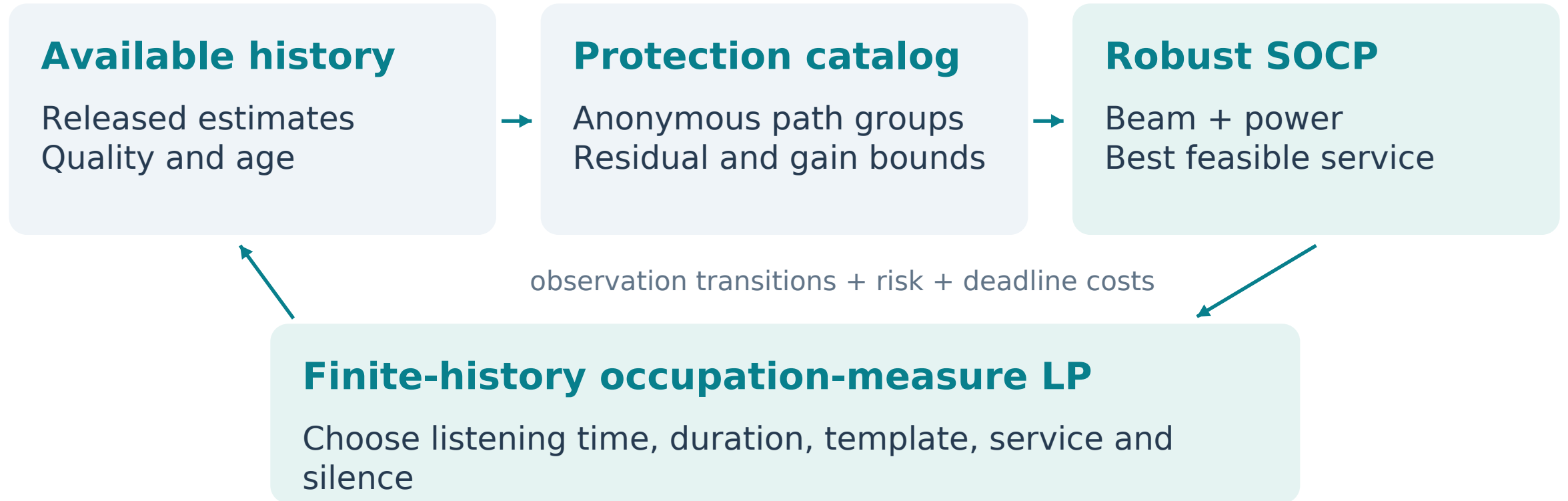
One controlled sector, two networks, partial information



Known online: current TN CSI, released observations, actions and service history.

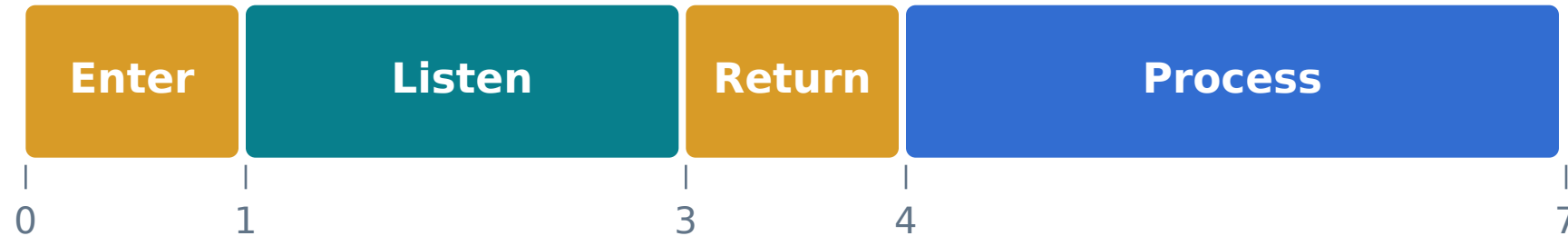
Hidden: true NTN DL channels, identities, future activity and future observations.

The appendix connects sensing to service through two optimizers



SOCP chooses the best protected transmission; LP decides when it is worth using.

A new estimate is unavailable until its release time



**RF unavailable: no affected TN
DL/UL**

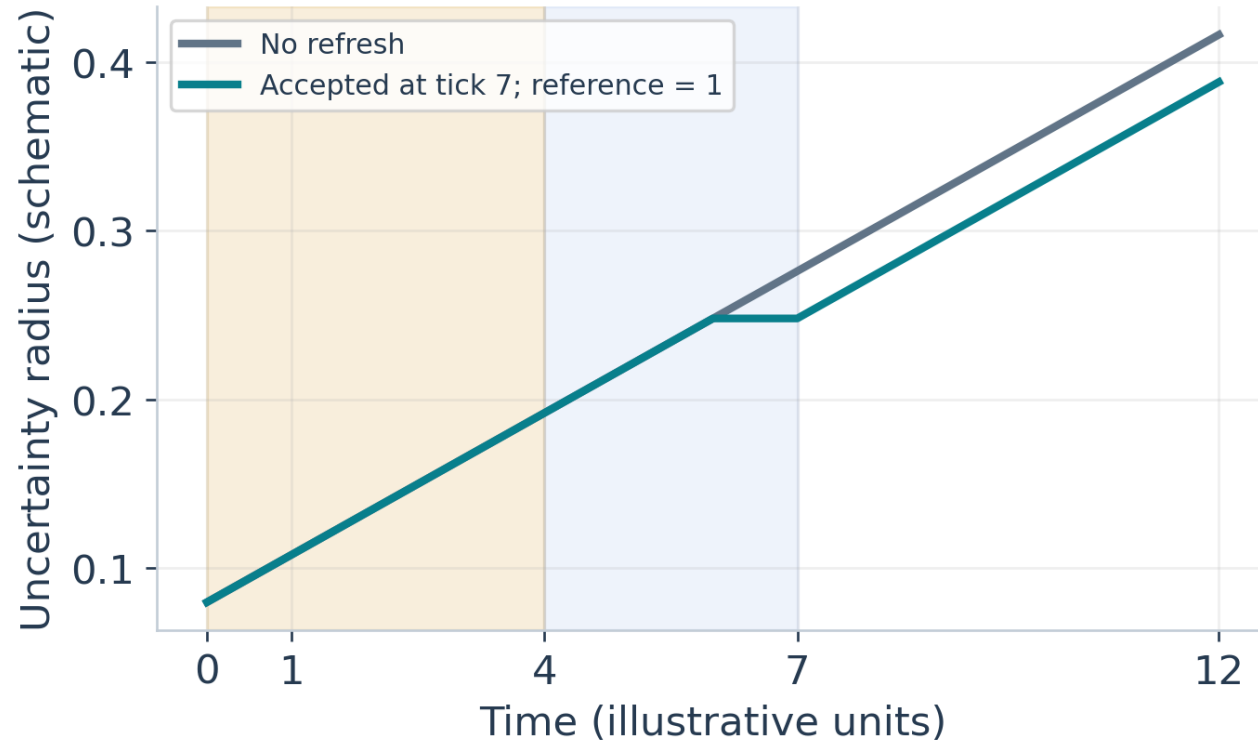
**RF returned: service may resume
using the old record**

$$t_{\text{ref}} = s + g^{\text{in}}, \quad t_{\text{use}} = s + g^{\text{in}} + \tau + g^{\text{out}} + t_p$$

$$\Delta_0 = \tau + g^{\text{out}} + t_p$$

At release: age = 7 – 1 = 6. A rejected observation keeps the previous record.

Longer listening trades accuracy against usable service time

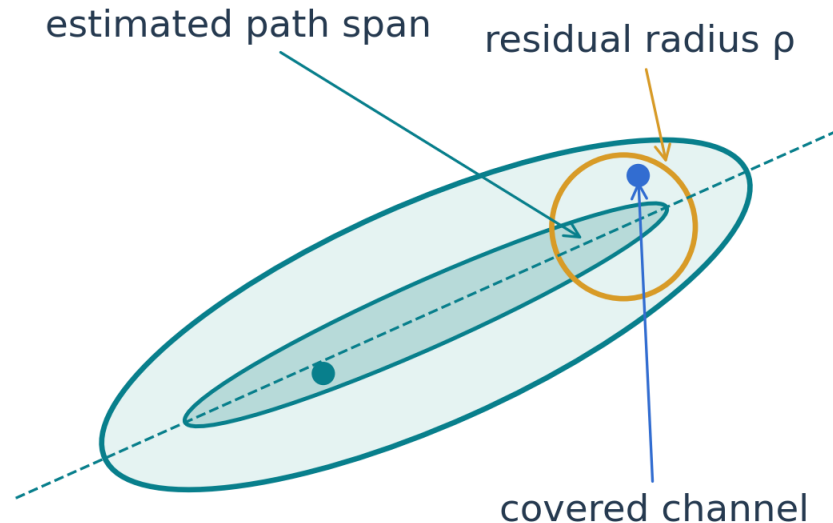


- Age grows from the sample reference time.
- An accepted refresh may shrink uncertainty.
- Longer sensing also delays release and consumes service time.

The policy must compare successful, rejected and no-detection branches.

Protect a set of complete channels, not individual peaks

$$\mathcal{G}_r = \{\mathbf{A}_r \mathbf{c} + \mathbf{e} : \|\mathbf{c}\|_2 \leq C_r, \|\mathbf{e}\|_2 \leq \rho_r\}$$



$\mathbf{A}_r$: reconstructed DL path responses

C_r : bound on complex coefficients

ρ_r : unresolved paths and model error

r : anonymous group, not a UE identity

Complex coefficients retain coherent path addition; the full receiver channel must be covered.

The worst-case leakage has an exact closed form

$$\sup_{\mathbf{f} \in \mathcal{G}_r} |\mathbf{f}^H \mathbf{v}| = C_r \|\mathbf{A}_r^H \mathbf{v}\|_2 + \rho_r \|\mathbf{v}\|_2$$

Modeled-path contribution

Suppress the estimated DL subspace.
More directions can cost spatial freedom.

Residual contribution

Uncertain directions remain.
Power backoff limits their leakage.

$$C_r \|\mathbf{A}_r^H \mathbf{v}\|_2 + \rho_r \|\mathbf{v}\|_2 \leq \sqrt{\Gamma}$$

Exact for the declared set; the set itself may conservatively overbound physical channels.

A silent receiver can still force power backoff

$$\mathbf{v} = \sqrt{p} \mathbf{w}, \quad \|\mathbf{w}\|_2 = 1, \quad \text{INR}_i = |\mathbf{f}_i^H \mathbf{v}|^2$$

$$\mathbf{f}_i = \sqrt{\chi_i P^{\max}/N_i} \mathbf{g}_i^D$$

Directional information

A known path span supports spatial suppression.

Unknown complex DL gains still require valid bounds.

Arbitrary-direction background

If $\|\mathbf{f}\| \leq \beta$, every direction is possible.

The only universal control is beam norm.

$$\beta \|\mathbf{v}\|_2 \leq \sqrt{\Gamma} \Rightarrow p \leq \min\{1, \Gamma/\beta^2\}$$

Passive UL sensing cannot distinguish an absent receiver from a permanently silent one.

Service reliability and interference exposure use different metrics

TN: deadline service

Accumulate only available resources.
Check each user, direction and window.
The demand is frozen in advance.

$$\Pr \{B_{u,x,m} < L_{u,x,m}\} \leq \delta_{u,x,m}^{\text{TN}}$$

Mean throughput cannot replace this event.

NTN: active-time exposure

Include missed and UL-silent receivers.
Count activity during gaps and silence.
Aggregate durations across episodes.

$$\mathcal{O}_{i,m} = \frac{\mathbb{E}[\int_{W_m} z_i(t) \mathbf{1}\{\text{INR}_i > \Gamma\} dt]}{\mathbb{E}[\int_{W_m} z_i(t) dt]}$$

A ratio of expectations, not a window event.

Use identical windows, calendars, references and budgets for every policy.

Coverage risk bridges robust constraints and actual outage

$$\Pr \{E_i(t) \mid \mathcal{H}_k, a, z_i(t) = 1\} \leq \eta_i$$

Coverage failure

Missing receivers
or missing groups

Set failure

Channel outside the
modeled uncertainty set

Gain-bound failure

Noise / gain / coupling
bounds do not hold

$$\eta_i = \min\{1, \eta_i^{\text{cov}} + \eta_i^{\text{set}} + \eta_i^{\text{gain}}\}, \quad c_{i,m} = \eta_i d_{i,m}$$

Marginal calibration coverage is not a certificate for every adaptively selected history.

The joint problem maximizes net service under explicit budgets

$$\max_{\Pi \text{ causal}} \mathbb{E}_{\Pi} \left[\sum_{u, x, m} B_{u, x, m} \right]$$

$$\Pr_{\Pi} \{B_{u, x, m} < L_{u, x, m}\} \leq \delta_{u, x, m}^{\text{TN}}$$

$$C_{i, m}(\Pi) \leq \delta_i^{\text{NTN}} D_{i, m}(\Pi)$$

Calendar • RF availability • transmit power • robust set constraints

The old leakage penalty λ is replaced by physical constraints and service requirements.

At one state, a convex program selects beam and power

$$V_k = \max_{\mathbf{v}} \operatorname{Re}\{\mathbf{h}^H \mathbf{v}\}$$

$$\operatorname{Im}\{\mathbf{h}^H \mathbf{v}\} = 0, \quad \|\mathbf{v}\|_2 \leq 1$$

$$C_r \|\mathbf{A}_r^H \mathbf{v}\|_2 + \rho_r \|\mathbf{v}\|_2 \leq \sqrt{\Gamma}$$

for every imposed protection group r

- Rotate the common phase: no loss of optimality.
- Recover $p = \|\mathbf{v}\|^2$ and the beam direction.
- Convert V_k into service; retain mute separately.

This solves the best current protected transmission—not the sensing schedule.

Why the outer controller does not search every beam

Coupling argument

Keep the same physical trajectory.
Keep sensing and mute decisions.
Replace each nonmute beam by the SOCP optimizer.

Delivered bits cannot decrease.

Conditions that make it valid

Fixed calendar and external interference.
No energy carryover or switching cost.
Exogenous physical/observation laws.
One shared certified nonmute risk bound.

The result concerns certified risk; beam-dependent true-outage costs need a different reduction.

Keep one SOCP maximizer and an explicit mute action at each available history.

A causal state contains what is known—not the hidden channel

S = time + revealed history + RF / pending status + delivered bits + record timestamps



Pending metadata includes the release time, but never the unreleased estimate.

Occupation measures turn policy optimization into an LP

$x(S,a)$ = probability of reaching state S and choosing action a

$$\max_{x \geq 0} \sum_{S,a} x(S,a) r(S,a)$$

$$\sum_a x(S,a) = \mu_0(S) + \sum_{\bar{S}, \bar{a}} P(S | \bar{S}, \bar{a}) x(\bar{S}, \bar{a})$$

$$\sum_{S,a} x(S,a) \bar{g}_{u,x,m}(S,a) \leq \delta_{u,x,m}^{\text{TN}}$$

$$\sum_{S,a} x(S,a) [c_{i,m}(S,a) - \delta_i^{\text{NTN}} d_{i,m}(S,a)] \leq 0$$

$$\pi^*(a | S) = x^*(S,a) / \sum_{a'} x^*(S,a')$$

Normalize state-action mass to recover the executable randomized policy.

Randomization is sometimes required, not merely convenient

Policy A (illustration)

NTN cost = 0.02
TN failure = 0.20

Policy B (illustration)

NTN cost = 0.08
TN failure = 0.02

50 / 50 mixture

NTN cost = 0.05
TN failure = 0.11

Example limits: NTN cost ≤ 0.05 and TN failure ≤ 0.11 .
A fails TN, B fails NTN; the mixture satisfies both.

Actual E6 check: 128 deterministic policies; none feasible in the randomization case; mixed optimum = 1.125.

Executing only the most likely action can destroy the probability constraints.

There are three distinct levels of guarantee

1 Set-wise protection

Every channel inside an imposed set satisfies the INR limit.

2 Finite-model optimality

Exact SOCP + complete finite history + exact transitions/costs → optimal certified policy.

3 Physical-system protection

Also requires valid physical state laws, active-time expectations and conditional certificates.

Most optimization is offline; execution follows released events

Freeze the model

Calendar, demands,
kernels and risk bounds



Build and solve

Reachable histories
SOCP values → LP



Execute causally

Sample an action
Reveal only released
results

Carry pending jobs, record ages and service counters through deadlines.
Charge online computation latency. Report infeasibility separately from solver failure.

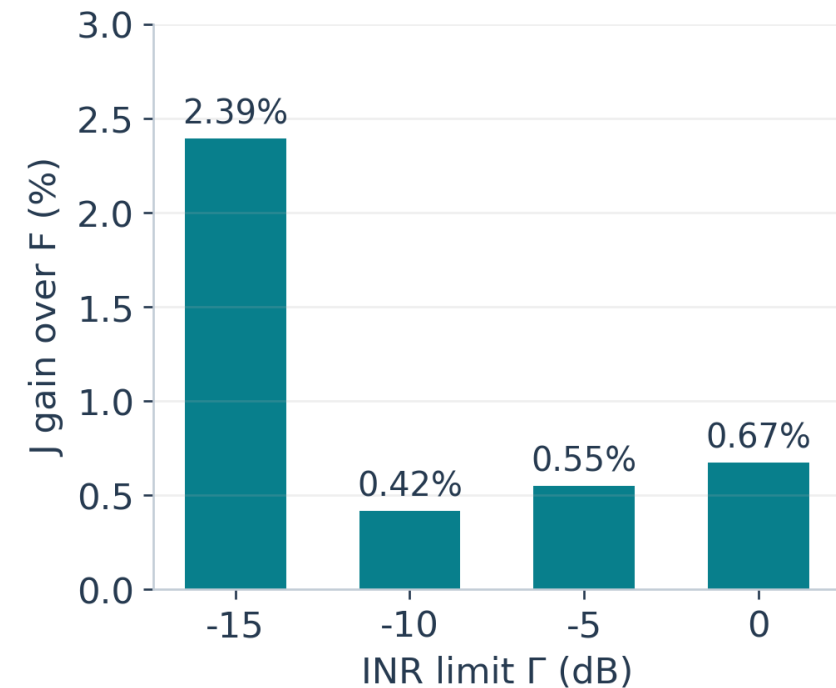
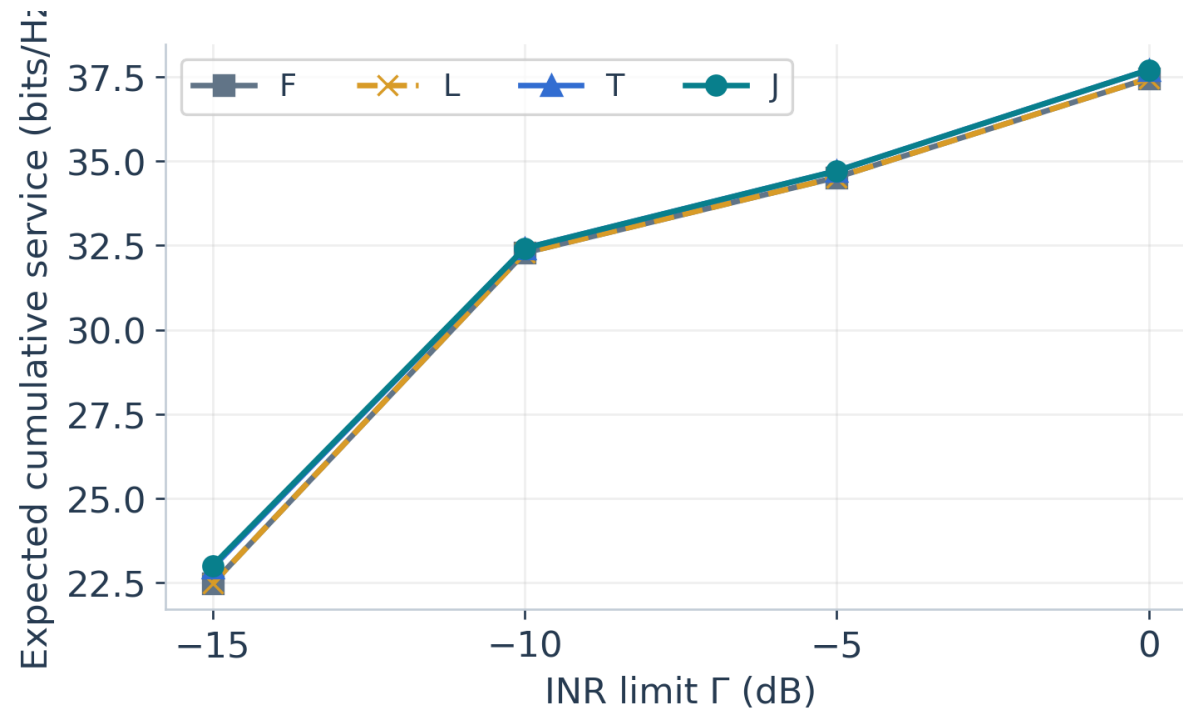
The complete history can be exponential even though the explicit LP is polynomial in size.

What the current experiments validate

E4	Set calibration and coverage	Synthetic whole-scene split
E5	Worst leakage and robust SOCP	Numerical identities + geometry tests
E6	Beam reduction and policy LP	Complete-policy enumeration
E7	F / T / L / J service tradeoff	Declared finite dynamic model
E8	Failure, delay, arrivals, correlation	Model stress + fixed causal diagnostics
Fresh RT	New drops + DL/UL tracing + Γ -CDF	Static held-out physical channels

Finite-model gains are positive, but small

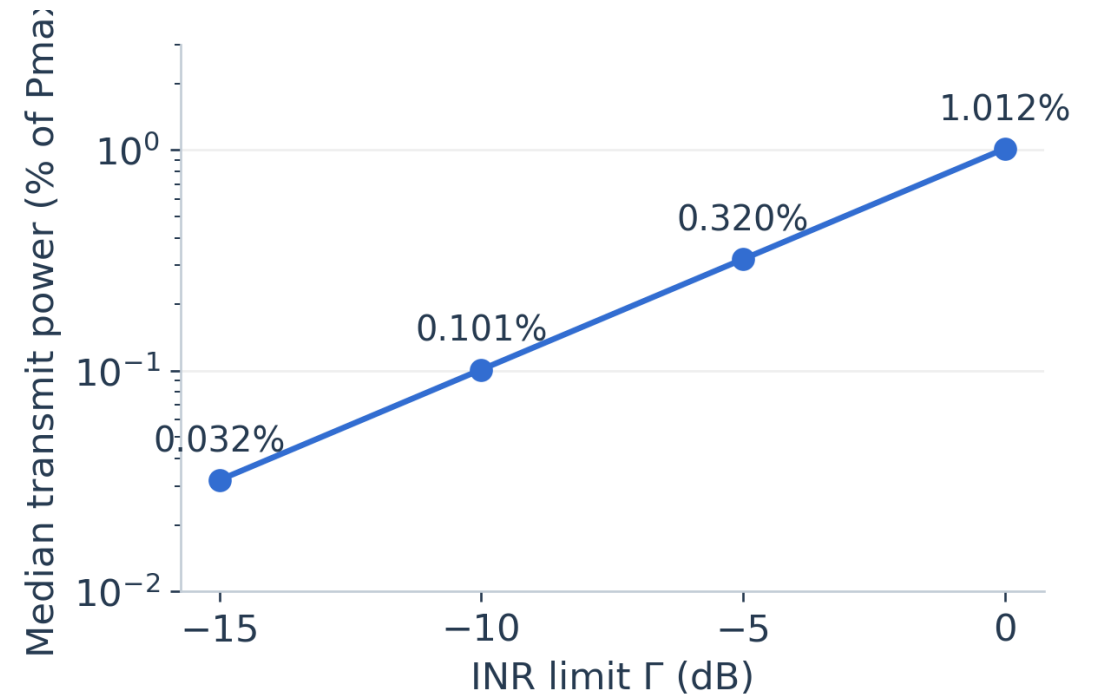
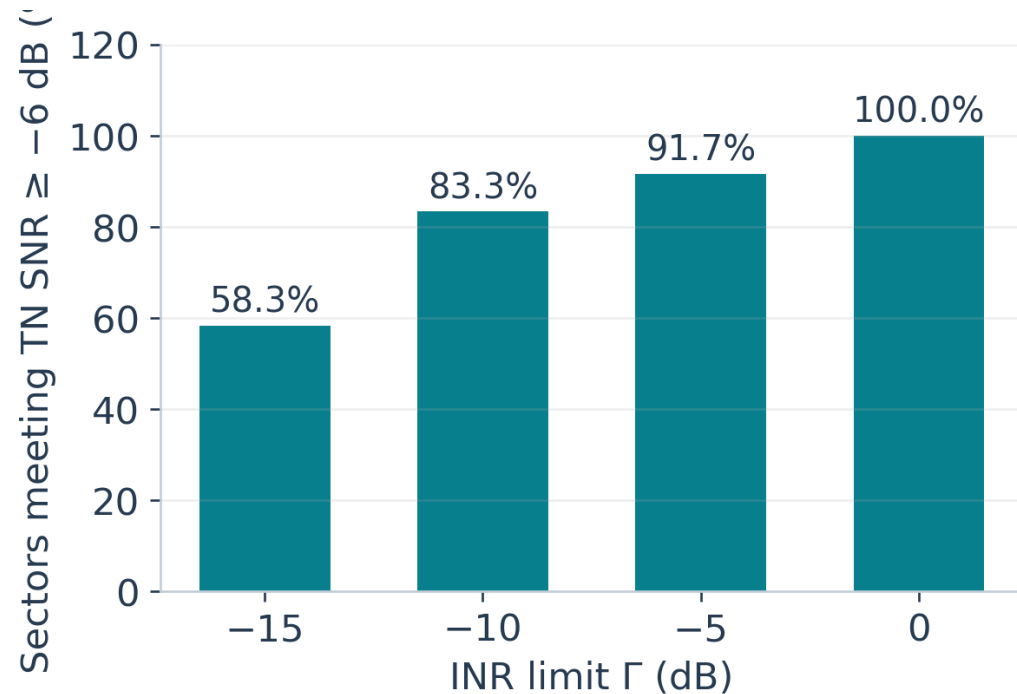
F: fixed T: timing only L: duration only J: joint adaptation



J = T at three Γ values; current evidence does not show a broad duration-adaptation gain.

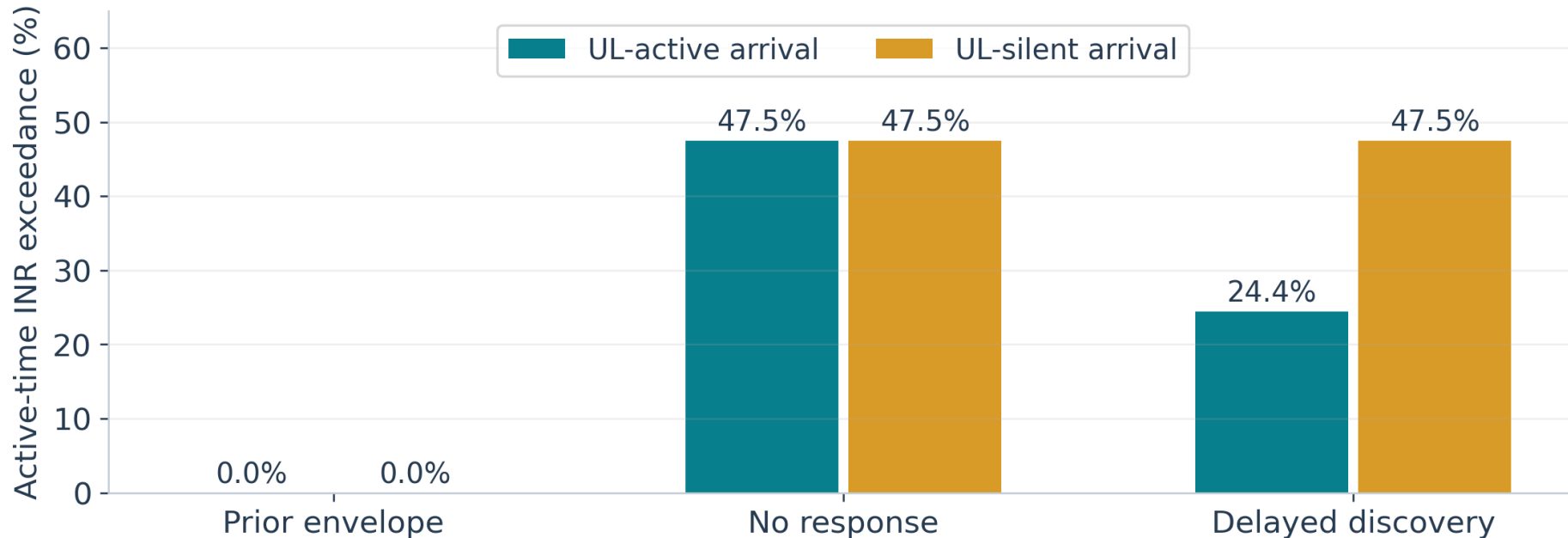
Fresh ray tracing reveals a substantial service cost

7 GHz DL / 6.3 GHz UL · 64 antennas · depth 3 · 800-snapshot MDL-MUSIC



Low INR alone is insufficient: the current empirical envelope is conservative.

Unknown arrivals expose the gap before usable information



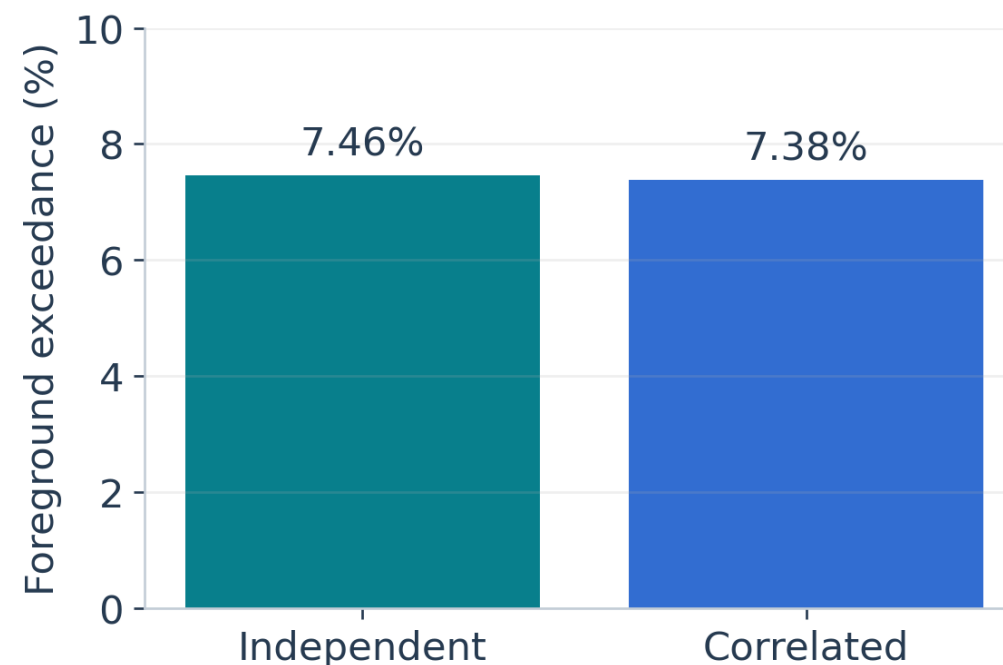
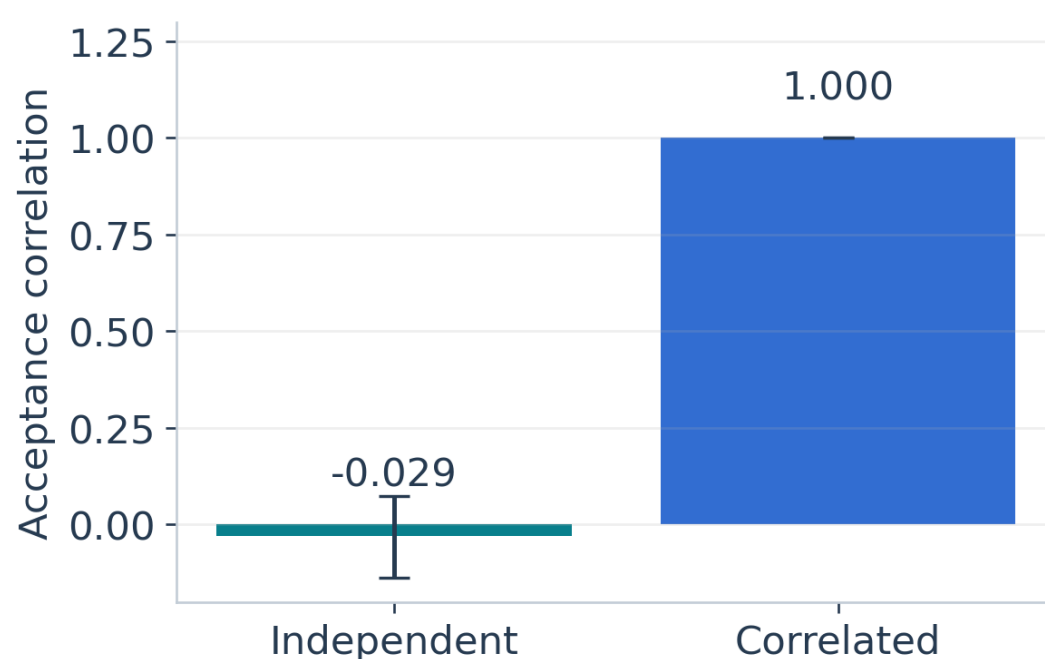
Only released detections reach the controller.

UL-active case: detection 67.5%; observed exceedance after protection activates = 0.

Fixed diagnostics violate the main TN QoS budget; these are not optimized J policies.

UL-silent arrivals require prior coverage; delayed discovery cannot protect the unseen past.

Repeated observations now genuinely test temporal correlation



Every path executes two listens. The fixed diagnostic violates the main TN budget; it tests the observation mechanism, not a feasible optimized-J service gain.

Correlation is present; this run does not show a statistically significant performance loss.

The contribution is a causal connection from sensing to service

One channel-set model

Single paths, coherent multipath and background uncertainty share a robust constraint.

One structured optimization

SOCP supplies protected service values; an occupation LP chooses a causal policy.

One clear validation boundary

Finite-model optimality is established conditionally; physical dynamic calibration remains to be built.

How listening quality enters the multipath residual

$$\mathbf{f}_i = (\mathbf{A}_r + \Delta \mathbf{A}_r) \mathbf{c} + \mathbf{h}_r^{\text{tail}}$$

$$\rho_r = C_r \left(\sum_p \varepsilon_{rp}^2 \right)^{1/2} + \rho_r^{\text{tail}}$$

$$\varepsilon(\tau, \Delta, \nu) = L[r_0(\tau, \nu) + \nu \Delta] + e_{\text{cal}} + e_{\text{mp}}$$

Finite observation error + directional aging + calibration / multipath mismatch.
 C_r requires path-power bounds or independent calibration: coherent cancellation can hide large path coefficients in a small total channel.

Raw UL MUSIC weights do not certify protected-band gain or noise bounds.

Silence is feasible; positive deadline service may not be

$$V_k \geq \sqrt{\frac{N+I}{p_{\max}} (2^{d/(BT_{\text{rem}})} - 1)}$$

d remaining bits; T_{rem} available DL time; fixed channel and catalog.

$$\ell_r = C_r \sqrt{\lambda_{\min}(\mathbf{A}_r \mathbf{A}_r^H)} + \rho_r$$

$$p_{\text{cap}} = \min\{1, \min_{r: \ell_r > 0} \Gamma / \ell_r^2\}, \quad V_k \leq \|\mathbf{h}\|_2 \sqrt{p_{\text{cap}}}$$

This is a constant-input screening bound, not a universal per-symbol QoS constraint.

A network extension needs local budgets and an external reserve

Sector 1

Local amplitude/risk
budgets

Sector 2

Local amplitude/risk
budgets

Sector 3

Local amplitude/risk
budgets

$$\sum_b \Gamma_b \leq \Gamma^{\text{net}}, \quad \sum_b \delta_{ib} \leq \delta_i^{\text{net}}$$

$$\Gamma^{\text{net}} = \Gamma^{\text{total}} - \bar{j}^{\text{ext}}$$

Independent data streams add average powers; failure events need not be independent.

A target, an estimate and a confidence bound are different

E4 scene miscoverage

Target: 10%

Observed: $13 / 128 = 10.16\%$

95% interval: 5.52%–16.74%

Compatible with target; not proof of a $\leq 10\%$ physical risk.

E7: J at $\Gamma = -15$ dB

Model risk bound: 8%

Observed exposure: 8.19%

95% bootstrap: 6.67%–9.75%

400 complete trajectories; ticks are not independent samples.

Split and resample whole scenes / trajectories. Keep missed and silent receivers.

Zero observed events do not establish zero risk.

Notation and source map

$\mathbf{v} = \sqrt{\mathbf{p}} \mathbf{w}$

Power-weighted transmit beam; actual power = $P_{\max} p$

$\mathbf{A}_r, \mathbf{C}_r, \rho_r$

DL path responses, coefficient bound, residual radius

$\Gamma / \delta \mathbf{N} \mathbf{T} \mathbf{N}$

INR magnitude limit / active-time exceedance budget

$\mathbf{L} / \delta \mathbf{T} \mathbf{N}$

Frozen deadline demand / service-failure budget

$\eta / \mathbf{c} / \mathbf{d}$

Conditional coverage-risk bound / risk time / active time

$\mathbf{S} / \mathbf{x}(\mathbf{S}, \mathbf{a}) / \pi(\mathbf{a} | \mathbf{S})$

Observable history state / occupation mass / action probability

Paper: [overleaf_ntn_paper/direction.tex](#)

Results: [result/appendix_d_20260921_235448_775259/](#)